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import java.util.Scanner;

public class MatrixMultiplication {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Get dimensions of first matrix
        System.out.print("Enter number of rows in first matrix: ");
        int rowsA = scanner.nextInt();
        System.out.print("Enter number of columns in first matrix: ");
        int colsA = scanner.nextInt();

        // Get dimensions of second matrix
        System.out.print("Enter number of rows in second matrix: ");
        int rowsB = scanner.nextInt();
        System.out.print("Enter number of columns in second matrix: ");
        int colsB = scanner.nextInt();

        // Check if compatible for multiplication
        if (colsA != rowsB) {
            System.err.println("Incompatible matrices for multiplication! Rows of first matrix must equal columns of second matrix.");
            System.exit(1);
        }

        // Create matrices
        int[][] matrixA = new int[rowsA][colsA];
        int[][] matrixB = new int[rowsB][colsB];
        int[][] productMatrix = new int[rowsA][colsB];

        // Get user input for matrices
        System.out.println("\nEnter elements of first matrix:");
        for (int i = 0; i < rowsA; i++) {
            for (int j = 0; j < colsA; j++) {
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        System.out.print "[" + i + ", " + j + "] = ";
        matrixA[i][j] = scanner.nextInt();
    }
}

System.out.println("\nEnter elements of second matrix:");
for (int i = 0; i < rowsB; i++) {
    for (int j = 0; j < colsB; j++) {
        System.out.print "[" + i + ", " + j + "] = ";
        matrixB[i][j] = scanner.nextInt();
    }
}

// Perform matrix multiplication
for (int i = 0; i < rowsA; i++) {
    for (int j = 0; j < colsB; j++) {
        productMatrix[i][j] = 0;
        for (int k = 0; k < colsA; k++) {
            productMatrix[i][j] += matrixA[i][k] * matrixB[k][j];
        }
    }
}

// Print product matrix
System.out.println("\nProduct of matrices:");
for (int i = 0; i < rowsA; i++) {
    for (int j = 0; j < colsB; j++) {
        System.out.print(productMatrix[i][j] + "\t");
    }
    System.out.println();
}
}
}

```